

Beyond First-Order Task Vector Superposition: Orthogonal Residuals in In-Context Learning

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Abstract

When in-context examples span multiple tasks, large language models (LLMs) can represent and perform them simultaneously, a phenomenon known as task superposition. A common hypothesis is that the resulting hidden representation is a convex combination of the individual task vectors. Whether this describes the hidden representations of instruction-tuned LLMs, and how faithfully it holds across depth, has not been directly examined. We find that the convex prediction accounts for the component of the true mixed-task representation that lies along it, at the right magnitude. This is tested across five instruction-tuned models, three task groups, and all probed layers, averaged across ratios. We then characterize the orthogonal residual that the prediction leaves unaccounted for and observe that its magnitude increases with depth. Finally, causal ablations in Llama-3.2-3B show that this residual is behaviorally meaningful: removing its systematic component changes the model’s next-token distribution more than matched random orthogonal perturbations as layer depth increases. Together, these results show that task superposition in instruction-tuned LLMs is accurately convex in its parallel component, with an additional orthogonal structure that emerges systematically across depth.

1 Introduction

In-context learning (ICL) enables large language models to adapt to new tasks from only a handful of demonstrations, without updating model parameters (Brown et al., 2020). Recent work has shown that in-context learning induces latent task representations, or task vectors, within transformer hidden states (Hendel et al., 2023; Dong et al., 2026). Building on this perspective, Xiong et al. (2025) introduced task superposition, hypothesizing that

mixed-task prompts produce representations that are linear combinations of individual task vectors. While this hypothesis is supported by behavioral evidence and visualization, a mechanistic explanation for task superposition at the representational level remains an open question.

Verifying these representations matters for understanding how transformers combine multiple tasks internally and for representation steering. Existing evidence for task superposition is largely behavioral, leaving open whether the convex-combination hypothesis fully describes the hidden representations of instruction-tuned models. We make three contributions:

- We directly test the convex-combination hypothesis at the representational level across three task groups and five instruction-tuned LLMs. Averaged across different mixed-task ratios, the predicted convex combination has the right magnitude along its own direction at every probed layer, across all five models and three task groups.
- We characterize the orthogonal residual that this prediction leaves unaccounted for. Although the residual grows with depth, whether it clears its sampling noise floor varies by task group, localizing where linear superposition is most faithful.
- We causally evaluate the orthogonal residual by ablating its systematic component during the forward pass. Removing this direction perturbs the next-token distribution more than norm-matched random orthogonal directions at deeper layers, showing that the residual encodes behaviorally meaningful information beyond the linear prediction.

Code and data are available at the repository https://github.com/Aman-Amith-Shastri/task_vector_superposition for reproducibility.

2 Related work

Task Vector Geometry in In-Context Learning

Recent work has characterized in-context learning through the lens of task vectors—latent vector representations in the transformer’s residual stream that summarize the task specified by a set of in-context demonstrations. Hendel et al. (2023) introduce the task vector $\theta(S)$, derived from in-context examples S , and show that it captures the task-relevant information necessary for in-context learning. Using a test question x , a function $f(x, \theta)$ closely reproduces the behavior of the original transformer conditioned on the demonstrations S , suggesting that task vectors provide a compact representation of the inferred task.

Dong et al. (2026) build on this perspective through *Task Vectors as Representative Demonstrations*, proposing that task vectors form compressed representations of demonstrations through linear combinations of the original examples. Extending this idea, Yan et al. (2026) study inference-time ICL in small transformers trained on synthetic latent-task distributions and show that hidden states form convex combinations of learned task vectors, with coefficients corresponding to the Bayesian posterior over latent tasks. This convex regime holds only for in-distribution tasks; they attribute out-of-distribution generalization to an additional orthogonal subspace beyond it, whereas our test questions remain strictly in-distribution, so the orthogonal residual we study arises even where their convex prediction is expected to hold exactly. We instead ask whether the same convex geometry holds empirically at the activation level in instruction-tuned models, where the mixing weights are fixed by the prompt rather than inferred.

While these works study representations induced by a single task, Xiong et al. (2025) introduce *task superposition*—the ability of LLMs to perform multiple tasks simultaneously. They establish this behavior through top- K gold-token probabilities and LDA visualizations of task vectors across model families and scales. We instead verify the convex-combination hypothesis directly at the activation level in instruction-tuned models. Behavioral superposition does not itself imply that hidden representations combine linearly, and our central object is the orthogonal residual to the predicted convex combination, tracked across depth relative to a sampling noise floor.

Supporting this depth-dependent picture, Jiang

et al. (2025) report that ICL representations become compact and discriminative in early-to-middle layers before later layers express them to incorporate the query. This offers a mechanism for the sharpening we observe in the pure-task displacements, and hence for the drop in the Entity group’s noise floor at middle depths (Section 4.2).

Notions of Superposition

Superposition carries two senses in deep learning: *feature* superposition, where neurons encode multiple unrelated features through representational compression (Elhage et al., 2022), and the *task* superposition of Xiong et al. (2025)—performing multiple tasks from a single in-context prompt—which is the sense we adopt throughout. The two sit at different levels of abstraction, but both describe multiple entities sharing a common latent representation.

3 Experimental setup

3.1 Task groups and in-context example ratios

We consider three task groups—Arithmetic addition, Entity attribution to countries, and multiple-choice questions from Massive Multitask Language Understanding (MMLU) (Hendrycks et al., 2021)—chosen so that each varies one source of variation while holding the others fixed: *question format*, *semantic relation*, and *task semantics* respectively.

Arithmetic varies the *format* of the in-context examples while preserving semantic content: two randomly generated integers $op1$ and $op2$ are presented as a direct sum, as a multiple-choice selection (A–D), or as a verification of a proposed sum. All three formats share the same A: answer trigger, isolating format differences from the output prompt. **Entity** varies the *semantic relation*: a country is mapped to its capital, its currency, or its language, using a manually constructed dataset; unlike the other groups, no assistant trigger is used in the user prompt. **MMLU** varies *task semantics* while holding question format constant: multiple-choice questions from the World History, Law, and Machine Learning domains share a common prompt format and the Answer: trigger, eliciting a single-letter response. Prompt templates and sampling implementation details are provided in Appendix E.

Every prompt contains 3 in-context examples and 1 test question, and we vary the tasks the examples are drawn from. For each task group we consider 10 ratios: the pure ratios (3, 0, 0), (0, 3, 0),

and $(0, 0, 3)$, where all in-context examples belong to the same task (e.g. World History for MMLU, direct addition for Arithmetic), and the mixed ratios $(2, 1, 0)$, $(1, 2, 0)$, $(0, 2, 1)$, $(0, 1, 2)$, $(2, 0, 1)$, $(1, 0, 2)$, which draw on two tasks, and $(1, 1, 1)$ taking one example from each.

3.2 Contrast vectors

The residual stream activations have been used to analyze correlations with task/function vectors. Hendel et al. (2023) demonstrated that in-context examples (which they refer to as S) are represented in a task vector $\theta(S)$, which serve as the parameters which act on the test question input x . Thus, the transformer output can be abstracted as $f(x; \theta(S))$. Injecting the extracted task vectors into the residual stream with a zero shot prompt with the same test question implicitly assumes a zero-shot baseline. Here, we make the zero-shot baseline explicit by defining a contrast vector.

Let $S = [(x_1, y_1), (x_2, y_2), (x_3, y_3)]$ be the in-context examples presented in input-output pairs. (x_i, y_i) are sampled from different tasks in a single task group according to the required task ratio r . As shown in the in-context examples in Section 3.1, the x_i questions are the user prompts and corresponding y_i answers are assistant messages. Let x be the test question. Thus, $p_r(x) := [S, x]$ be the prompt with in-context examples and $p(x)$ is the corresponding zero-shot prompt with the same test question.

Define $\mathbf{h}_l(p_r(x))$ and $\mathbf{h}_l(p(x))$ as the activation at the final token position in layer l with prompts $p_r(x)$ (few shot with task ratio r and test question x) and $p(x)$ (corresponding zero shot prompt) respectively. Thus, we define the contrast vector $\mathbf{c}_r^l(x)$ as

$$\mathbf{c}_r^l(x) := \mathbf{h}_l(p_r(x)) - \mathbf{h}_l(p(x))$$

For each ratio r , we sample n test questions and ICL examples to generate n contrast vectors at a layer l . In the following sections, we will refer to the i^{th} test question as x_i . For each model, we collect contrast vectors at increasing layer depths proportional to the number of layers in the model.

3.3 Centroids and displacements

We quantify the extent to which contrast vectors (and by extension, task vectors) superpose with a metric based on a scalar projection. For each ratio

r , the centroid of the contrast vectors at layer l is

$$\mu_r^l := \frac{1}{n} \sum_{i=1}^n \mathbf{c}_r^l(x_i)$$

The convex-combination hypothesis posits that if k ICL examples are drawn for a single prompt from multiple tasks in a ratio r , the resulting contrast vector at layer l is a convex combination of the pure task centroids, weighted by the fraction of each task in the ICL examples—the weights r_p/k are non-negative and sum to one. Formally, for a ratio $(r_1, r_2, \dots, r_{|P|})$, the predicted centroid of the contrast vectors for ratio r and layer l is

$$\hat{\mu}_r^l := \sum_{p \in P} \frac{r_p}{k} \mu_p^l$$

where r_p denotes the component of ratio r corresponding to pure ratio $p \in P$. Let R be the set of all 10 ratios and P the set of pure ratios $(3, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 3)$ (Section 3.1), whose in-context examples are sampled from a single task. The tri-center of contrast vectors at layer l is

$$\mu^l := \frac{1}{|P|} \sum_{r \in R} \mu_r^l [[r \in P]]$$

From μ_r^l , $\hat{\mu}_r^l$, and μ^l we define the displacement vectors $\mathbf{d}_r^l := \mu_r^l - \mu^l$ and $\hat{\mathbf{d}}_r^l := \hat{\mu}_r^l - \mu^l$; Figure 1 shows their relationships in 2D.

3.4 Metrics

There are four relevant metrics that we analyze in our experiments:

- κ : the scalar coefficient of the projection of the actual displacement vector $\mathbf{d}_r^l$ onto the predicted displacement vector $\hat{\mathbf{d}}_r^l$
- s_r^l : the orthogonal residual
- ϕ_r^l : the sampling noise floor, and
- $\tilde{s}_r^l$: the size-matched residual

3.4.1 Scalar projection of $\mathbf{d}_r^l$ onto $\hat{\mathbf{d}}_r^l$

From $\mathbf{d}$ and $\hat{\mathbf{d}}$, we can now define κ :

$$\kappa_r^l := \frac{\mathbf{d}_r^l \cdot \hat{\mathbf{d}}_r^l}{\|\hat{\mathbf{d}}_r^l\|^2}$$

κ is the factor by which $\hat{\mathbf{d}}_r^l$ must be scaled to match the projection of $\mathbf{d}_r^l$ onto it. So $\kappa = 1$ means

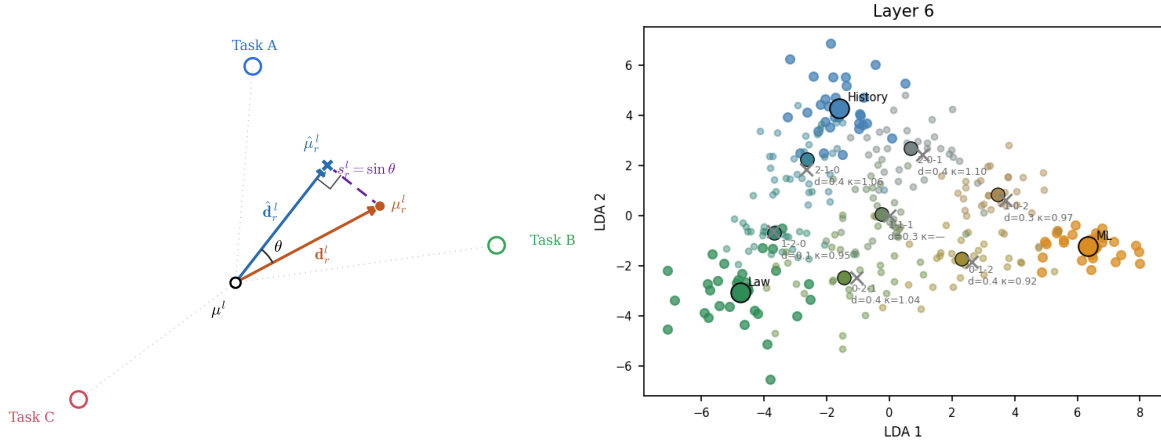


Figure 1: Geometry of the convex-combination prediction. (a) Relationship between the tri-center, pure-task centroids, and the observed versus predicted mixed-ratio centroids and contrast vectors. (b) An LDA projection of contrast vectors for the MMLU task group in Llama 3.2 3B (layer 6), showing predicted versus actual centroids for a mixed ratio. **Colors:** Blue = History, Orange = ML, Green = Law. **Markers:** $\times$ predicted centroid, $\bullet$ actual centroid for a ratio.

the prediction has the right magnitude along its own direction; it says nothing about any orthogonal component, while $\kappa = 0$ indicates complete orthogonality.

Note that for $r = (1, 1, 1)$, $\hat{\mu}_{(1,1,1)}^l = \frac{1}{3}\mu_{(3,0,0)}^l + \frac{1}{3}\mu_{(0,3,0)}^l + \frac{1}{3}\mu_{(0,0,3)}^l = \mu^l$: the predicted centroid and the tri-center coincide, so $\hat{\mathbf{d}}_{1-1-1}^l = \mathbf{0}$ and κ_{1-1-1}^l is undefined. We therefore do not compute κ for the 1-1-1 ratio.

3.4.2 Orthogonal residual

κ pins the component of the observed displacement *along* the predicted convex direction, but is blind to any *orthogonal* component: a centroid could project onto the prediction with unit coefficient while still lying off the predicted point. Thus, for a given prompt ratio r at layer l , we measure the orthogonal residual s_r^l as

$$s_r^l := \sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l) = \sqrt{1 - \frac{(\mathbf{d}_r^l \cdot \hat{\mathbf{d}}_r^l)^2}{\|\mathbf{d}_r^l\|^2 \|\hat{\mathbf{d}}_r^l\|^2}} \quad (1)$$

3.4.3 Noise floor

To establish the extent to which the residual s_r^l is providing a clear signal, we calibrate it against a sampling noise floor ϕ_r^l generated by randomly splitting the n samples of contrast vectors per ratio and layer in half $B = 500$ times. For a split b , we calculate the displacement vectors $\mathbf{d}^{(1,b)}$ and $\mathbf{d}^{(2,b)}$ from the disjoint splits of size $n/2$ according to the ratio r : $\mathbf{d}^{(1,b)} = \frac{2}{n} \sum_{i=1}^{n/2} \mathbf{c}_r^l(x_i) - \mu^l$ and $\mathbf{d}^{(2,b)} = \frac{2}{n} \sum_{i=n/2+1}^n \mathbf{c}_r^l(x_i) - \mu^l$. Formally, for

ratio r at layer l we define the sampling noise floor ϕ_r^l and the size-matched residual $\tilde{s}_r^l$ as

$$\phi_r^l := \frac{1}{B} \sum_{b=1}^B \sin \theta(\mathbf{d}^{(1,b)}, \mathbf{d}^{(2,b)}), \quad (2)$$

$$\tilde{s}_r^l := \frac{1}{2B} \sum_{b=1}^B \sum_{j \in \{1,2\}} \sin \theta(\mathbf{d}^{(j,b)}, \hat{\mathbf{d}}_r^l) \quad (3)$$

The size-matched residual $\tilde{s}_r^l$ uses $n/2$ samples per displacement vector, matching that of the floor. It is an average over the displacement vectors across all $2B$ splits, where each $\mathbf{d}^{(j,b)}$ is calculated from $n/2$ samples in split b .

3.5 Ablating the orthogonal residual

To test whether the residual’s direction matters to the output, we remove it from the residual stream and measure the effect on the model’s output.

Stage 1: extract the residual. For each mixed ratio r and layer l we first partition the n samples of contrast vectors into a fitted group and a held-out group. We then average the contrast vectors over the fitted group to get its mean contrast $\bar{\mathbf{c}}_r^l$, and keep the part of its displacement from the tri-center μ^l orthogonal to the convex prediction $\hat{\mathbf{d}}_r^l$:

$$\mathbf{v}_\perp(r) := (\bar{\mathbf{c}}_r^l - \mu^l) - \kappa(r) \hat{\mathbf{d}}_r^l \perp \hat{\mathbf{d}}_r^l,$$

where $\kappa(r)$ is the scalar projection of the displacement onto $\hat{\mathbf{d}}_r^l$. So $\mathbf{v}_\perp(r)$ is the average orthogonal residual per (ratio, layer). Averaging over individual contrast vectors cancels the idiosyncrasy of any

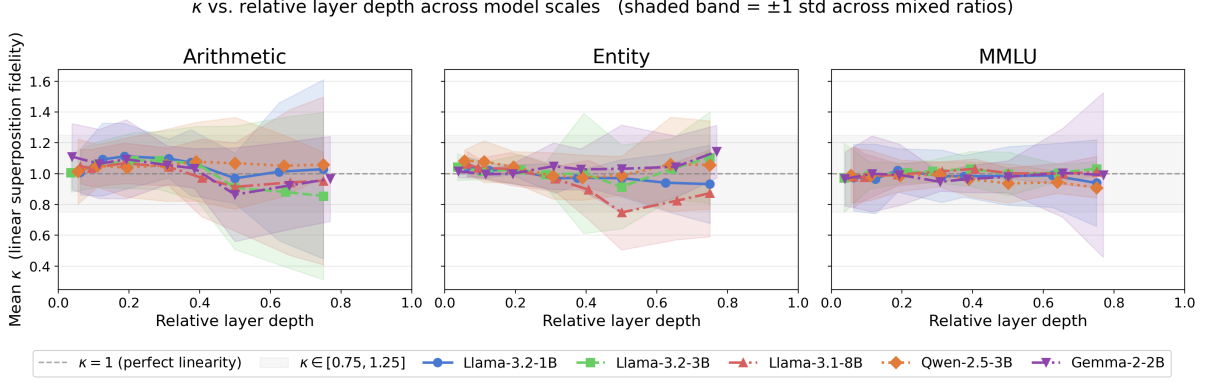


Figure 2: Cross-model κ versus relative layer depth for the Arithmetic, Entity, and MMLU task groups (panels share a y -axis). Each line is one model’s mean κ across the $K = 6$ mixed ratios; the band of the same color is ± 1 standard deviation of κ across those ratios, a dispersion across ratios rather than an uncertainty on the mean. The dashed line marks $\kappa = 1$ (exact linear superposition) and the grey zone $\kappa \in [0.75, 1.25]$.

individual prompt. Further, keeping the fitted and held-out groups disjoint also helps avoid any bias in the ablation test.

Stage 2: remove it. For each question in the held-out group we subtract $\mathbf{v}_\perp(r)$ from that question’s own few-shot activation at the same ratio r and the same layer l , at the final token position. The null arm substitutes one of 30 random vectors $\mathbf{g}$, each also orthogonal to $\hat{\mathbf{d}}_r^l$ and rescaled so that $\|\mathbf{g}\| = \|\mathbf{v}_\perp(r)\|$. Both arms remove a single out-of-sample vector from the few-shot residual stream, and differ only in its direction.

Stage 3: measure the shift. We report $\text{KL}(\text{few-shot} \parallel \text{ablated})$ – the divergence between the unmodified and ablated next-token distributions – averaged over the held-out group. We also report the rank/ p -value of the ablation of the original residual amongst the $D = 30$ random orthogonal norm-matched vectors. Because the two arms are matched in norm, in orthogonality, and in the prompts and questions they act on, any gap between them is attributable to direction alone. The question split, construction details, the faithfulness control, and per-layer values are deferred to Appendix D.

4 Results

4.1 The convex combination accounts for the parallel component of the observed displacement

We extract contrast vectors for the Arithmetic, Entity, and MMLU task groups from five instruction-tuned models spanning multiple families and scales: Llama 3.2 1B, Llama 3.2 3B, Llama 3.1 8B (Grattafiori et al., 2024), Gemma 2 2B (Gemma

Team, 2024), and Qwen 2.5 3B (Yang et al., 2024). Following Xiong et al. (2025), we visualize them with an LDA projection fit on the pure-task centroids (Figure 1), extending their analysis of task superposition from pre-trained models to practically deployed instruction-tuned LLMs. Implementation details for prompt construction, relative layer selection, contrast vector extraction, and κ computation are in Appendix A; representative layer-wise visualizations are in Appendix F, with the complete set for every model and task group in the accompanying code repository.

Across all five models, all three task groups, and every sampled layer, κ remains centered close to 1 (Figure 2). Against the exact permutation null of Appendix B.1, which re-pairs each observed displacement with another ratio’s predicted convex combination and is centered at 0, the observed mean κ attains the smallest p the null admits, $p = 1/720$, in all 120 model-group-layer cells.

The mean stays near 1 at every depth, but the spread across mixed ratios widens in later layers (the standard deviation across ratios rises by 0.25 for Arithmetic, 0.19 for Entity, and 0.024 for MMLU between the first and last sampled layers). To investigate whether this observation is simply due to increased interactions between activations throughout the forward pass or potentially another “signal” containing task-specific information, we turn to the orthogonal residual.

4.2 The orthogonal residual grows with relative layer depth

The orthogonal residual s_r^l grows with depth in every task group, from $\sin \theta \approx 0.30$ – 0.42 at the shallowest swept layer to ≈ 0.53 – 0.83 at the deep-

Group	First layer			Last layer			Within floor
	full	half	floor	full	half	floor	
Arithmetic	0.38	0.45	0.47	0.76	0.81	0.75	7/40
Entity	0.34	0.68	0.94	0.67	0.69	0.45	21/40
MMLU	0.37	0.57	0.80	0.70	0.89	0.92	37/40

Table 1: Orthogonal-residual calibration via split-half consistency, at the *first* and *last* swept layer of each model, averaged over the mixed ratios and then over all five models. “full”: the *full-sample* orthogonal residual $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$, the minimum-variance estimate of its magnitude (0 means the mixed centroid lies exactly on the predicted point). “half”: the *size-matched* residual, a half-sample estimate at the floor’s own $n/2$ sample size. “floor”: the sampling noise floor ($n/2$). “Within floor” pools, over the five models, the swept layers (8 per model, 40 in total) whose size-matched residual is at or below the floor, i.e. indistinguishable from an exact convex combination given sampling noise. Group means can conceal real spread across models; per-model values are given in Appendix C.2 (Table 5).

est (Appendix C). Averaged across models, the increase is monotonic for MMLU and rises before leveling off in the Arithmetic and Entity groups. Linear superposition is therefore geometrically tightest in the early layers, where the residual lies within the sampling noise floor in all but two of the fifteen model–task combinations. Table 1 and Figure 3 report the full-sample and size-matched residuals, as well as the floors across all 5 models, 3 groups, and 8 layer depths.

The residual has different relations to the corresponding noise floors in each task group. For Arithmetic, the residual increases faster than the floor and clears it in the earlier layers. By contrast, the Entity group’s floor reduces and the residual increases most drastically in the middle layers. Finally, MMLU’s floor remains high and its residual remains near the floor throughout, exceeding it only marginally at the deepest swept layers in Gemma-2B and Qwen-3B. We conjecture that this ordering tracks how separable the pure-task vectors are: MMLU’s three subjects share a common multiple-choice format and single-letter output, which would keep their centroids close together and the floor correspondingly high, while Arithmetic’s three formats differ in surface form.

Although the residual is shown to increase with relative depth, it is unclear whether it actually is an artifact of task vector superposition. We thus turn to the ablation experiment in Section 3.5 and analyze whether the residual carries some task-specific information.

4.3 Removing the orthogonal residual changes the output more than a norm-matched random direction

The effect depends on the residual’s direction.

For Llama-3.2-3B at the deepest layer we probe (layer 21, three-quarter depth), removing $\mathbf{v}_\perp(r)$ shifts the next-token distribution more than every one of the 30 norm-matched random orthogonal draws, in all six mixed ratios, in all three task groups. Pooled over ratios, the ablation reaches 0.044 nats against a null mean of 0.005 for Arithmetic— $8.0\times$ its largest draw—0.297 against 0.042 for Entity ($4.7\times$ its largest draw), and 0.0039 against 0.0004 for MMLU ($6.2\times$ its largest draw). Removing this particular orthogonal direction therefore costs more than removing any other orthogonal direction of the same length.

The ratio grows with relative layer depth. For all three task groups, we see an increasing gap between the effects of the true residual ablation and random norm-matched orthogonal vectors on the model outputs. At the shallowest probed layer, the ablation lands wherever a random direction of the same length would: its percentile within the 30 null draws scatters from the 0th to the 87th across ratios. However, from layer 11 onward the true ablation’s effect exceeds that of every null draw (Table 6). Both arms grow with depth, which is expected — an intervention closer to the output layer leaves less remaining computation to absorb it. However, the ratio of these effects grows as well; for Arithmetic the true ablation has $1.5\times$ the effect on model outputs compared to the null mean at relative depth 0.21 and $9.9\times$ at relative depth 0.75. Hence, the growth is not explained by proximity to the output alone. The trend is not monotonic (as seen in the Entity task group), but its direction is consistent across all three task groups.

MMLU: detectability and causal effect dissociate.

MMLU’s orthogonal residual stays within its sampling noise floor at 37 of the 40 swept layers (Section 4.2), clearing it only marginally at the deepest layers of Gemma-2B and Qwen-3B. However, for Llama-3.2-3B at layers 11–21, removing its average residual shifts the next-token distribution more than every null draw in every mixed ratio. The two tests measure different properties. The sampling noise floor tests whether the residual’s size can be distinguished from sampling noise in the displacement; the ablation asks whether remov-

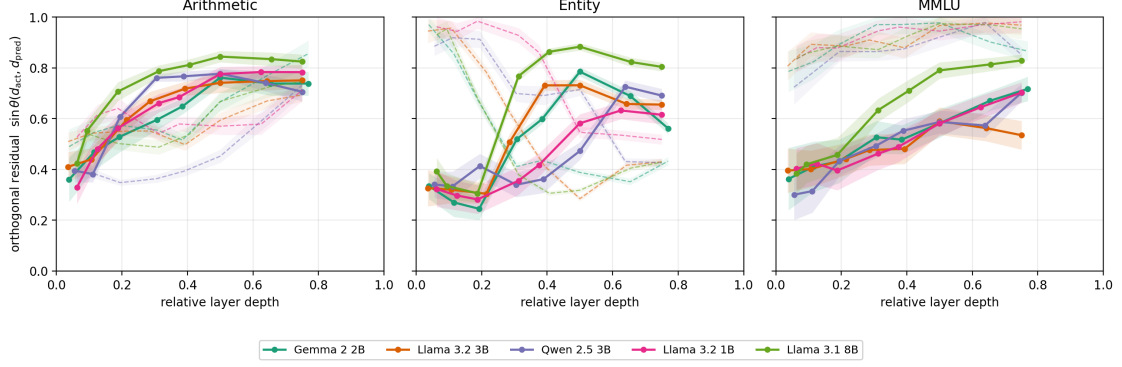


Figure 3: Orthogonal residual of the convex-combination prediction versus relative layer depth, one panel per task group. Solid lines show the full-sample orthogonal residual $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$; dashed lines show the corresponding sampling noise floor ($n/2$).

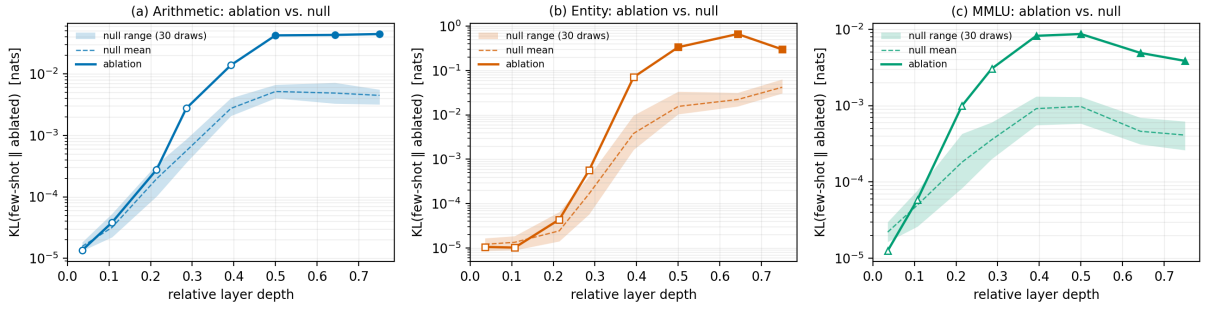


Figure 4: Ablating the average orthogonal residual at the final token position across ratios (Llama-3.2-3B), against relative layer depth, one panel per task group: **(a)** Arithmetic, **(b)** Entity, **(c)** MMLU. Each shows the KL divergence of the ablated next-token distribution from the few-shot distribution (solid), against the norm-matched random-orthogonal null (dashed: null mean; band: full range of the $D = 30$ draws). Filled markers denote layers where the ablation exceeds every null draw in every mixed ratio; hollow markers denote layers where it does not. Points are means over the 10-question held-out group and 6 mixed ratios, with centroids fit on the 40-question fitted group (20 for MMLU). Note the log scale, and that the three panels use different y -ranges. Per-layer values are given in Table 6.

ing the fitted residual’s direction changes the output more than removing a random orthogonal direction of the same length. A residual can therefore be too noisy to resolve geometrically while still pointing in a task-relevant direction.

5 Conclusion

We analyze the proposed convex combination of pure-task centroids and find that κ remains centered close to 1 across five models, three task groups, and every sampled layer: the prediction accounts for the component of the observed displacement along its own direction. It is blind to the orthogonal component, which we measure as the residual s_r^l . The residual grows with relative layer depth in every task group, and whether it clears the sampling noise floor varies by task group. For Llama-3.2-3B at layers 11–21, removing the fitted residual shifts the next-token distribution more than every one of the 30 norm-matched random orthogonal

draws, in all six mixed ratios, while at the shallowest probed layer it is indistinguishable from those draws. Thus, a convex combination of task representations captures the first-order term, but leaves a residual whose direction the output depends on at these depths.

Limitations

Depth coverage. Our layer sweep extends only to a relative depth of 0.75, leaving the final $\sim 25\%$ of layers unprobed. This is consistent with the emerging picture of how depth is used: the last stage of inference is a “residual sharpening” phase that suppresses features to finalize the next-token distribution rather than building the representations we analyze (Lad et al., 2025), and intermediate rather than final layers carry the richest, most linearly structured task representations (Skean et al., 2024). The task-composition geometry we study is therefore expected to be established by this depth;

we nonetheless do not verify that the linear structure persists—or degrades in the manner these accounts predict—across the terminal output-oriented layers, and leave a targeted sweep of that band to future work.

Positional encoding. We do not directly control for positional encoding, although we are able to demonstrate linearity without explicitly controlling for this factor. Given that the prompt lengths for the three task groups vary significantly, positional encoding could have measurable effects that are unaccounted for by the convex model for task vector superposition we use.

Model coverage. All five models we test are instruction-tuned. [Xiong et al. \(2025\)](#) establish task superposition in pre-trained models, and we do not test whether the geometry we report differs in base models of the same family.

Hardware constraints. Model scale is capped at 8B parameters due to hardware constraints, so whether the same depth- and separability-dependent picture holds at larger scales is left open. Further, residual-ablation intervention (Section 4.3) is run on a single model, Llama-3.2-3B, picked arbitrarily among those tested due to hardware constraints; whether the same causal picture holds at other model families and scales is left open.

AI Assistance

Claude Code was used to aid experiment design, coding, data analysis, literature search, and writing and editing. All AI-assisted code, analyses, and text were reviewed by the author, who takes full responsibility for the content.

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A Implementation details for empirical verification

A.1 Prompt construction

For each of the 10 ratios listed in Section 3.1, n samples of in-context examples and test questions are created for each task group. $n = 50$ for the Arithmetic and Entity task groups and $n = 30$ for MMLU tasks.

For arithmetic tasks, op1 and op2 are sampled from the range [1, 49] and are formatted as Direct, MCQ, or Verification tasks. For entity tasks, countries are sampled and their corresponding entities are extracted based on the ratio. For MMLU tasks, questions for a sample are extracted from each task (subject/domain) based on the ratio.

For a given task group and ratio, each sample constitutes (1) the in context examples concatenated with the test question to produce the few shot prompt, and (2) the corresponding test question, used as the zero-shot baseline. For all three task groups, the test question is extracted from a heldout pool disjoint from the in-context examples. Implementation details on sampling the ICL examples and test question can be found in Appendix E.

A.2 Layer selection

We use the following relative layer depths for each model: [0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.65, 0.75]. The higher density in earlier layers is intended to capture the region where task-relevant information is most concentrated, motivated by findings that task and function vectors are most effective when injected in the early-to-middle layers of autoregressive LLMs (Hendel et al., 2023; Todd et al., 2024).

A.3 Contrast vector extraction

For each sample, we first collect the residual stream activations at the final token position. Since these activations are independent of the task ratio in in-context examples, they are precomputed and stored in a temporary cache for efficiency. This is done by rotating the task corresponding to the test question for each sample. That is, for sample index i , the test question is drawn from the task $\text{TASKS}[i \% \text{num_tasks}]$.

Next, for each ratio, we conduct another forward pass to collect the activations at the final token position when given the few shot prompts and the corresponding test questions used in the zero shot prompts.

The stored zero shot activations are then subtracted from the corresponding few shot activations to produce contrast vectors at each layer of the model for each ratio. For visualizing these vectors, following the method used by Xiong et al. (2025), we use LDA projection. LDA is fit on the pure ratio contrast vectors only ((3, 0, 0), (0, 3, 0), and (0, 0, 3)). This is then used to transform the contrast vectors from all 10 ratios. A separate LDA is fit per layer per task group.

A.4 Calculating κ

The above contrast vectors are then used to compute μ_r^l , $\hat{\mu}_r^l$, and μ^l as described in Section 3.3 for each layer at each ratio. The calculated values of κ are then stored per model and task group. κ is calculated using the original contrast vectors, not the LDA projected vectors, providing a measurement of linearity in the original activation space.

Further, as stated in Section 3.4.1, κ is not well defined for the 1-1-1 ratio and is therefore not computed.

B Calibration and aggregate statistics for κ

B.1 Exact permutation null

To calibrate the observed κ values, we use an exact permutation null. κ is defined on the $K = 6$ mixed ratios whose predicted displacement $\hat{\mathbf{d}}_r^l$ is nonzero (the three pure ratios define the basis, and 1-1-1 is undefined). The observed statistic at layer l is the mean $S := \frac{1}{K} \sum_r \kappa_r^l$. A permutation π re-pairs each observed displacement $\mathbf{d}_r^l$ with a permuted predicted displacement $\hat{\mathbf{d}}_{\pi(r)}^l$, yielding

$$S(\pi) := \frac{1}{K} \sum_r \frac{\mathbf{d}_r^l \cdot \hat{\mathbf{d}}_{\pi(r)}^l}{\|\hat{\mathbf{d}}_{\pi(r)}^l\|^2}$$

We enumerate all $6! = 720$ permutations and report the one-sided $p = \#\{\pi : S(\pi) \geq S\}/720$. The identity permutation is included, so the smallest attainable value is $p = 1/720 \approx 1.4 \times 10^{-3}$; this is the exact-enumeration analogue of the add-one correction (Phipson and Smyth, 2010). The null is computed per layer and stored alongside κ for each model and task group.

B.2 Aggregate statistics across ratios and depth

Tables 2–4 report the full κ distribution behind the near-constant summary quoted in the main text. For each task group–model pair and each swept layer l (relative depth $d = l/L$), we give the mean $\bar{\kappa}$, standard deviation σ_{κ} , and range $[\min, \max]$ of κ_r^l over the $K = 6$ mixed ratios. The per-layer mean stays close to 1 at every depth in every cell, indicating that the prediction is scale-accurate along its own direction throughout; the dispersion across ratios widens somewhat at the deepest layers—most visibly in the Arithmetic group, where individual ratios spread while the mean holds near 1—consistent with the growing orthogonal residual analyzed in Section 4.2 and Appendix C.

Table 2: Aggregate κ statistics for the Arithmetic task group: per-layer mean, standard deviation, and range across the $K = 6$ mixed ratios, for each model. Relative depth is $d = l/L$.

Model	l (d)	$\bar{\kappa}$	σ_{κ}	range
Llama 3.2 1B	1 (0.06)	1.01	0.16	[0.76, 1.25]
	2 (0.12)	1.09	0.26	[0.68, 1.42]
	3 (0.19)	1.11	0.23	[0.73, 1.33]
	5 (0.31)	1.10	0.14	[0.99, 1.30]
	6 (0.38)	1.07	0.23	[0.72, 1.33]
	8 (0.50)	0.97	0.18	[0.83, 1.31]
	10 (0.62)	1.01	0.49	[0.35, 1.81]
	12 (0.75)	1.03	0.64	[0.15, 1.94]
Gemma 2 2B	1 (0.04)	1.11	0.24	[0.77, 1.40]
	3 (0.12)	1.06	0.25	[0.64, 1.26]
	5 (0.19)	1.09	0.28	[0.64, 1.44]
	8 (0.31)	1.06	0.14	[0.80, 1.16]
	10 (0.38)	1.03	0.14	[0.80, 1.18]
	13 (0.50)	0.86	0.33	[0.36, 1.20]
	17 (0.65)	0.92	0.31	[0.64, 1.36]
	20 (0.77)	0.97	0.30	[0.73, 1.42]
Llama 3.2 3B	1 (0.04)	1.01	0.14	[0.76, 1.17]
	3 (0.11)	1.06	0.16	[0.80, 1.22]
	6 (0.21)	1.08	0.20	[0.83, 1.39]
	8 (0.29)	1.09	0.21	[0.85, 1.36]
	11 (0.39)	1.07	0.26	[0.76, 1.37]
	14 (0.50)	0.91	0.44	[0.33, 1.44]
	18 (0.64)	0.88	0.53	[0.01, 1.41]
	21 (0.75)	0.86	0.60	[-0.18, 1.43]
Qwen 2.5 3B	2 (0.06)	1.01	0.23	[0.63, 1.31]
	4 (0.11)	1.05	0.15	[0.80, 1.22]
	7 (0.19)	1.04	0.21	[0.68, 1.20]
	11 (0.31)	1.05	0.26	[0.72, 1.35]
	14 (0.39)	1.08	0.28	[0.81, 1.57]
	18 (0.50)	1.07	0.33	[0.75, 1.63]
	23 (0.64)	1.05	0.22	[0.74, 1.40]
	27 (0.75)	1.06	0.09	[0.92, 1.14]
Llama 3.1 8B	2 (0.06)	1.05	0.12	[0.83, 1.14]
	3 (0.09)	1.04	0.13	[0.82, 1.14]
	6 (0.19)	1.07	0.14	[0.88, 1.29]
	10 (0.31)	1.04	0.13	[0.88, 1.22]
	13 (0.41)	0.97	0.28	[0.74, 1.32]
	16 (0.50)	0.91	0.31	[0.45, 1.28]
	21 (0.66)	0.94	0.52	[0.28, 1.50]
	24 (0.75)	0.95	0.60	[0.23, 1.61]

Table 3: Aggregate κ statistics for the Entity task group: per-layer mean, standard deviation, and range across the $K = 6$ mixed ratios, for each model. Relative depth is $d = l/L$.

Model	l (d)	$\bar{\kappa}$	σ_{κ}	range
Llama 3.2 1B	1 (0.06)	1.04	0.07	[0.94, 1.12]
	2 (0.12)	1.02	0.05	[0.94, 1.10]
	3 (0.19)	1.02	0.04	[0.96, 1.06]
	5 (0.31)	0.97	0.11	[0.84, 1.08]
	6 (0.38)	0.97	0.05	[0.90, 1.02]
	8 (0.50)	0.97	0.13	[0.73, 1.10]
	10 (0.62)	0.94	0.20	[0.68, 1.17]
	12 (0.75)	0.93	0.28	[0.61, 1.22]
Gemma 2 2B	1 (0.04)	1.02	0.05	[0.93, 1.06]
	3 (0.12)	1.00	0.03	[0.96, 1.03]
	5 (0.19)	1.00	0.07	[0.89, 1.08]
	8 (0.31)	1.05	0.17	[0.81, 1.22]
	10 (0.38)	1.03	0.15	[0.88, 1.26]
	13 (0.50)	1.03	0.31	[0.60, 1.43]
	17 (0.65)	1.04	0.22	[0.80, 1.32]
	20 (0.77)	1.14	0.19	[0.87, 1.30]
Llama 3.2 3B	1 (0.04)	1.04	0.09	[0.90, 1.15]
	3 (0.11)	1.02	0.06	[0.95, 1.11]
	6 (0.21)	1.03	0.11	[0.87, 1.14]
	8 (0.29)	0.99	0.09	[0.90, 1.13]
	11 (0.39)	1.00	0.43	[0.20, 1.37]
	14 (0.50)	0.91	0.30	[0.51, 1.30]
	18 (0.64)	1.03	0.22	[0.70, 1.27]
	21 (0.75)	1.10	0.32	[0.61, 1.47]
Qwen 2.5 3B	2 (0.06)	1.08	0.08	[0.99, 1.21]
	4 (0.11)	1.08	0.15	[0.84, 1.26]
	7 (0.19)	1.04	0.11	[0.85, 1.15]
	11 (0.31)	0.99	0.16	[0.78, 1.22]
	14 (0.39)	0.97	0.16	[0.72, 1.19]
	18 (0.50)	0.99	0.14	[0.88, 1.24]
	23 (0.64)	1.06	0.34	[0.63, 1.52]
	27 (0.75)	1.06	0.31	[0.57, 1.55]
Llama 3.1 8B	2 (0.06)	1.06	0.09	[0.93, 1.18]
	3 (0.09)	1.04	0.07	[0.96, 1.13]
	6 (0.19)	1.03	0.12	[0.84, 1.17]
	10 (0.31)	0.97	0.09	[0.90, 1.12]
	13 (0.41)	0.89	0.21	[0.64, 1.22]
	16 (0.50)	0.75	0.27	[0.36, 1.04]
	21 (0.66)	0.82	0.28	[0.53, 1.14]
	24 (0.75)	0.87	0.31	[0.51, 1.31]

Table 4: Aggregate κ statistics for the MMLU task group: per-layer mean, standard deviation, and range across the $K = 6$ mixed ratios, for each model. Relative depth is $d = l/L$.

Model	l (d)	$\bar{\kappa}$	σ_{κ}	range
Llama 3.2 1B	1 (0.06)	0.97	0.24	[0.68, 1.36]
	2 (0.12)	0.97	0.25	[0.75, 1.39]
	3 (0.19)	1.02	0.15	[0.79, 1.19]
	5 (0.31)	0.98	0.19	[0.72, 1.20]
	6 (0.38)	0.98	0.15	[0.75, 1.16]
	8 (0.50)	0.98	0.21	[0.72, 1.18]
	10 (0.62)	0.99	0.23	[0.70, 1.19]
	12 (0.75)	0.94	0.31	[0.56, 1.22]
Gemma 2 2B	1 (0.04)	0.97	0.19	[0.65, 1.21]
	3 (0.12)	1.00	0.27	[0.65, 1.46]
	5 (0.19)	0.99	0.22	[0.71, 1.37]
	8 (0.31)	0.95	0.07	[0.84, 1.01]
	10 (0.38)	0.96	0.10	[0.82, 1.07]
	13 (0.50)	0.98	0.11	[0.80, 1.12]
	17 (0.65)	1.01	0.31	[0.50, 1.32]
	20 (0.77)	0.99	0.59	[0.14, 1.63]
Llama 3.2 3B	1 (0.04)	0.97	0.25	[0.65, 1.38]
	3 (0.11)	0.99	0.10	[0.91, 1.15]
	6 (0.21)	1.01	0.07	[0.92, 1.10]
	8 (0.29)	1.02	0.08	[0.90, 1.11]
	11 (0.39)	1.03	0.12	[0.90, 1.23]
	14 (0.50)	1.00	0.15	[0.79, 1.22]
	18 (0.64)	1.00	0.06	[0.92, 1.08]
	21 (0.75)	1.03	0.05	[0.99, 1.11]
Qwen 2.5 3B	2 (0.06)	0.99	0.20	[0.71, 1.29]
	4 (0.11)	0.99	0.18	[0.77, 1.28]
	7 (0.19)	0.99	0.11	[0.81, 1.11]
	11 (0.31)	0.99	0.09	[0.83, 1.07]
	14 (0.39)	0.96	0.12	[0.75, 1.06]
	18 (0.50)	0.94	0.14	[0.76, 1.09]
	23 (0.64)	0.94	0.08	[0.81, 1.01]
	27 (0.75)	0.91	0.07	[0.78, 0.98]
Llama 3.1 8B	2 (0.06)	0.98	0.22	[0.69, 1.32]
	3 (0.09)	0.98	0.16	[0.79, 1.25]
	6 (0.19)	0.99	0.08	[0.93, 1.12]
	10 (0.31)	1.01	0.07	[0.93, 1.10]
	13 (0.41)	1.03	0.07	[0.94, 1.12]
	16 (0.50)	1.00	0.04	[0.93, 1.04]
	21 (0.66)	1.00	0.08	[0.87, 1.10]
	24 (0.75)	1.01	0.11	[0.88, 1.13]

C Sampling noise floor and per-model residual calibration

C.1 Construction of the floor

κ measures the component of the observed displacement $\mathbf{d}_r^l$ that lies *along* the predicted displacement $\hat{\mathbf{d}}_r^l$; the complementary *orthogonal* residual is captured by $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$, the fraction of $\mathbf{d}_r^l$ pointing off the predicted direction ($\sin \theta = 0$ iff the mixed centroid lies exactly on the predicted point). In finite samples this residual cannot reach 0, since $\mathbf{d}_r^l$ is an empirical centroid carrying sampling noise; comparing it to 0 is therefore uninformative. We instead calibrate it against the best alignment the data can resolve. For each mixed ratio r we randomly partition its n samples into two halves and form a displacement $\mathbf{d}^{(1)}, \mathbf{d}^{(2)}$ from each half, holding the pure-task centroids—and hence the triangle center and the prediction basis—at their full-sample values so that only the sampling noise in μ_r varies. The noise floor is $\sin \theta(\mathbf{d}^{(1)}, \mathbf{d}^{(2)})$, averaged over 500 random splits: no prediction can be expected to align with $\mathbf{d}_r^l$ more closely than an independent estimate of $\mathbf{d}_r^l$ aligns with itself. This yields two residual estimates that serve two distinct purposes. To report the *magnitude* of the orthogonal component and its trend across depth, we use the full-sample residual $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$ computed from all n samples: it is the minimum-variance estimate, so it tracks the true magnitude most cleanly and is not flattened by sampling noise (this is the quantity plotted in Figure 3 and reported in the “full” columns of Tables 1 and 5). To *test against the floor*—whether the residual exceeds what sampling noise alone would produce—we instead form a size-matched residual

on the same $n/2$ half-sample displacements as the floor, since these angles shrink with sample size and only an equal- n comparison is fair. The reported “within floor” flag is this size-matched residual falling at or below the floor: a residual within the floor indicates the orthogonal component is not distinguishable from sampling noise at this sample size, while one above it indicates an orthogonal component larger than sampling noise alone can explain. Reporting the full-sample residual against the $n/2$ floor would be a more lenient, best-case comparison, so we reserve the floor strictly for the size-matched test. The floor is a normalized, scale-free quantity, so it is comparable across layers, models, and task groups despite differing activation magnitudes.

C.2 Per-model values

Table 5 gives the per-model breakdown behind the group means of Table 1, with each group’s mean repeated in italics for reference. The group-level ordering—Arithmetic’s residual above the floor almost everywhere, Entity’s intermediate, MMLU’s within the floor at nearly every layer—is never reversed in an individual model, and is strict in four of the five; the exception is Gemma 2 2B, where Arithmetic and Entity tie at 3/8 within-floor layers. The means of Table 1 are therefore not artifacts of averaging over a heterogeneous set. The spread the means conceal is nonetheless real and is concentrated in two places. The deepest-layer floor varies substantially within the Arithmetic group (0.69–0.86), which is what drives the differing within-floor counts there (0/8 for Qwen 2.5 3B and Llama 3.1 8B, against 3/8 for Gemma 2 2B); and MMLU’s deepest-layer full-sample residual ranges from 0.53 (Llama 3.2 3B) to 0.83 (Llama 3.1 8B) around a mean of 0.70. Entity is the most homogeneous group, its deepest-layer floor sitting at 0.43 in four of the five models.

Group	Model	First layer			Last layer			Within floor
		full	half	floor	full	half	floor	
Arithmetic	Llama 3.2 1B	0.33	0.43	0.53	0.78	0.81	0.71	2/8
	Gemma 2 2B	0.36	0.44	0.49	0.74	0.81	0.86	3/8
	Llama 3.2 3B	0.41	0.49	0.51	0.75	0.79	0.69	2/8
	Qwen 2.5 3B	0.40	0.44	0.39	0.71	0.76	0.73	0/8
	Llama 3.1 8B	0.42	0.47	0.43	0.82	0.85	0.75	0/8
	<i>Mean</i>	<i>0.38</i>	<i>0.45</i>	<i>0.47</i>	<i>0.76</i>	<i>0.81</i>	<i>0.75</i>	<i>7/40</i>
Entity	Llama 3.2 1B	0.32	0.70	0.96	0.62	0.65	0.52	5/8
	Gemma 2 2B	0.34	0.76	0.97	0.56	0.59	0.43	3/8
	Llama 3.2 3B	0.33	0.65	0.94	0.66	0.67	0.43	4/8
	Qwen 2.5 3B	0.34	0.61	0.89	0.69	0.71	0.43	6/8
	Llama 3.1 8B	0.39	0.66	0.93	0.80	0.81	0.43	3/8
	<i>Mean</i>	<i>0.34</i>	<i>0.68</i>	<i>0.94</i>	<i>0.67</i>	<i>0.69</i>	<i>0.45</i>	<i>21/40</i>
MMLU	Llama 3.2 1B	0.40	0.61	0.83	0.70	0.87	0.98	8/8
	Gemma 2 2B	0.36	0.56	0.79	0.72	0.91	0.87	6/8
	Llama 3.2 3B	0.40	0.59	0.81	0.53	0.80	0.97	8/8
	Qwen 2.5 3B	0.30	0.48	0.72	0.70	0.93	0.85	7/8
	Llama 3.1 8B	0.38	0.59	0.83	0.83	0.93	0.95	8/8
	<i>Mean</i>	<i>0.37</i>	<i>0.57</i>	<i>0.80</i>	<i>0.70</i>	<i>0.89</i>	<i>0.92</i>	<i>37/40</i>

Table 5: Orthogonal-residual calibration via split-half consistency, per model and task group—the full breakdown summarized in Table 1. Columns are as in Table 1: for the *first* and *last* swept layer, averaged over the mixed ratios, the full-sample residual $\sin \theta(\mathbf{d}_r^l, \hat{\mathbf{d}}_r^l)$ (“full”), the size-matched $n/2$ residual (“half”), and the sampling noise floor (“floor”). “Within floor” counts the swept layers (of 8) whose size-matched residual is at or below the floor. Italicized *Mean* rows are the across-model means reported in Table 1; the within-floor mean is pooled over all 40 model–layer pairs rather than averaged.

D Residual ablation details

Decomposition. For a mixed ratio r at layer l , let $\bar{\mathbf{c}}_r^l$ be the mean of the contrast vectors $\mathbf{c}_r^l(x_i)$ over the fitted group and μ^l the tri-center (the mean of the three pure-ratio mean contrasts). Writing $\mathbf{d} = \bar{\mathbf{c}}_r^l - \mu^l$ for the displacement and $\hat{\mathbf{d}}_r^l$ for the convex prediction’s direction, the projection coefficient and the residual

Layer	Depth	Arithmetic			Entity			MMLU		
		ρ	KL	null	ρ	KL	null	ρ	KL	null
1	0.04	0.011	0.0000	0.0000	0.003	0.0000	0.0000	0.012	0.0000	0.0000
3	0.11	0.033	0.0000	0.0000	0.006	0.0000	0.0000	0.025	0.0001	0.0000
6	0.21	0.068	0.0003	0.0002	0.012	0.0000	0.0000	0.043	0.0010	0.0002
8	0.29	0.098	0.0028	0.0006	0.036	0.0006	0.0002	0.055	0.0030	0.0004
11	0.39	0.142	0.0138	0.0027	0.116	0.0705	0.0038	0.071	0.0082*	0.0009
14	0.50	0.169	0.0422*	0.0052	0.252	0.3336*	0.0155	0.069	0.0087*	0.0010
18	0.64	0.169	0.0429*	0.0049	0.229	0.6604*	0.0221	0.065	0.0049*	0.0005
21	0.75	0.162	0.0444*	0.0045	0.242	0.2974*	0.0418	0.067	0.0039*	0.0004

Table 6: Per-layer average residual magnitude $\rho = \|\mathbf{v}_\perp(r)\|/\|\mathbf{h}\|$, ablation divergence KL(few-shot $\parallel$ ablated) (nats), and the mean of the $D = 30$ norm-matched random-orthogonal null, for Llama-3.2-3B, averaged over the 10-question held-out group and the 6 mixed ratios, with centroids fit on the 40-question fitted group (20 for MMLU, whose pool is smaller). *: the ablation exceeds all 30 null draws in every mixed ratio at that layer ($p \leq 1/31 \approx 0.032$ per cell).

of Section 4.3 are

$$\kappa(r) = \frac{\langle \mathbf{d}, \hat{\mathbf{d}}_r^l \rangle}{\langle \hat{\mathbf{d}}_r^l, \hat{\mathbf{d}}_r^l \rangle}, \quad \mathbf{v}_\perp(r) = \mathbf{d} - \kappa(r) \hat{\mathbf{d}}_r^l,$$

so that $\mathbf{v}_\perp(r) \perp \hat{\mathbf{d}}_r^l$ by construction.

Faithfulness control. Overwriting the final-token state with the intact $\mathbf{h}_l(p_r(x_i))$ is a no-op: across all layers, ratios, and held-out questions the patched logits match the clean few-shot logits with $\max|\Delta\text{logit}| = 0$ and KL = 0, verifying that the patching mechanism itself introduces no artifact. All divergences reported are therefore attributable to the removal of $\mathbf{v}_\perp(r)$.

Null construction. Each $\mathbf{g}$ is drawn i.i.d. from $\mathcal{N}(0, I)$, projected onto the orthogonal complement of $\hat{\mathbf{d}}_r^l$, normalized, and rescaled to $\|\mathbf{v}_\perp(r)\|$ —so the null inherits its displacement magnitude from the fitted residual. The realized alignment with the true residual is negligible ($|\cos(\mathbf{g}, \mathbf{v}_\perp(r))| = 0.013\text{--}0.015$), confirming the draws do not accidentally recover the ablated direction. The D draws for a given (r, l) are evaluated as one batched forward pass over the shared prompt; batch rows do not interact, so this is arithmetically identical to D separate passes.

Protocol. We use a held-out group of 10 questions per ratio and the six non-degenerate mixed ratios (the balanced 1-1-1 ratio is excluded as degenerate: its convex prediction coincides with the tri-center, so $\hat{\mathbf{d}}$ is undefined), at layers taken at fixed fractions of model depth. Each ratio’s question pool is split disjointly by index: questions 0–9 form the held-out group on which the intervention is evaluated, and the remainder form the fitted group, from which every fitted quantity—pure-ratio mean contrasts, tri-center, and the mixed-ratio mean contrast—is estimated, so no sample belongs to both groups. Because test questions are drawn with replacement, index-disjointness does not imply content-disjointness: in Arithmetic, 2 of the 10 held-out questions recur among the 40 in the fitted group, each carrying weight 1/40 in the mean contrast; Entity has no recurrence. Arithmetic and Entity draw on a pool of 50 questions per ratio, leaving 40 for the fitted group; MMLU’s pool is 30, leaving 20, so its $\mathbf{v}_\perp(r)$ is the noisiest of the three. The ambient norm $\|\mathbf{h}\|$ is that of the actual few-shot final-token activation. Both ρ and κ are reported as means over the six mixed ratios. Per-layer values are given in Table 6.

E Implementation details for task group sampling

E.1 Prompt templates

Table 7 contains the formats for each task within the three groups, as well as examples of an in-context question-answer pair and a test question.

E.2 Arithmetic

First, a pool of 200 in-context and 50 test pairs (op1, op2), with each op1, op2 $\in [1, 49]$, are generated using seeds 42 and 43 respectively. Duplicate pairs are re-generated until the ICL and test pools are

Table 7: Prompt formats for each task and task group.

Group	Task	In-context Example	Test Question
Arithmetic	Direct	User: {op1} + {op2} A: Assistant: {op1+op2}	User: {op1} + {op2} A:
	MCQ	User: {op1}+{op2}=? A) {optA} B) {optB} C) {optC} D) {optD} A: Assistant: B	User: {op1}+{op2}=? A) {optA} B) {optB} C) {optC} D) {optD} A:
	Verification	User: {op1}+{op2}=x. Correct? A: Assistant: yes/no	User: {op1}+{op2}=x. Correct? A:
Entity	Capital	User: {country} Assistant: capital[{country}]	User: {country}
	Currency	User: {country} Assistant: currency[{country}]	User: {country}
	Language	User: {country} Assistant: language[{country}]	User: {country}
MMLU	History/Law/ML	User: Question: {question} A) {optA} B) {optB} C) {optC} D) {optD} Answer: Assistant: B	User: Question: {question} A) {optA} B) {optB} C) {optC} D) {optD} Answer:

disjoint. Thus, while the same (op1, op2) pair can appear in more than 1 in-context example, the in-context and test pools are disjoint. For each sample using ratio r of task combinations, 3 in-context op pairs and 1 test op pair are drawn without replacement. Each of the ICL pairs are formatted according to the ratio r and are concatenated with the test op pair.

Generating distractors for multiple choice: the distractors are within 10 of the true value of $op1 + op2$. This is to prevent lexical differences in the options having an unintended influence on the intended task to represent. For each ratio r , we use $n = 50$ samples.

E.3 Entity

We manually curate a dataset of 96 countries and map each to its capital, currency, and language. This dataset is partitioned into an in-context pool of 66 countries and a test pool of 30 countries, making the two pools disjoint.

To generate the in-context examples for a prompt, we sample countries without replacement from the in-context pool for each task according to the ratio r . This guarantees that countries are not repeated for a single task (that is, France $\rightarrow$ Euro does not appear twice in one sample prompt), but allows for cross-task duplication of countries (that is, France $\rightarrow$ Euro and France $\rightarrow$ Paris can appear in the same prompt). The test question is separately drawn from the test pool.

We use $n = 50$ samples for each in-context ratio. We do not add an assistant trigger in the user prompt after the test question; the prompt format is consistent across tasks and ratios.

E.4 MMLU

First, the questions from each subject (History, Law, ML) are shuffled by a seed (42). We use the test split of the MMLU dataset to avoid prior exposure of the train set questions to influence the model’s behavior. Next, the questions for each subject are partitioned into pools of up to 150 in-context questions and 20 test questions. Since ML has 112 questions in the MMLU dataset, we save 20 questions for the test pool and use the remaining 92 questions as the in-context pool. This guarantees the in-context and test pools are disjoint.

For each ratio r of task combinations, $n = 30$ samples are independently constructed, drawing 3 questions from the in-context pool according to r and one test set question. We use fewer samples here than for the arithmetic and entity groups for two concrete reasons. First, the MMLU subjects have

small question pools—ML contains only 112 questions, of which 92 remain for the in-context pool after reserving 20 for testing—so keeping n small bounds how often the same questions are reused across the independently constructed samples: at $n = 30$, the pure-MMLU ratio draws at most $3 \times 30 = 90$ in-context questions, within the 92-question ML pool, whereas $n = 50$ would force each question to be reused roughly $1.6\times$ on average, inflating correlation between samples. Second, MMLU prompts are full four-option questions with long stems, making each activation-extraction pass substantially more memory- and compute-intensive than the short arithmetic and entity prompts. We find $n = 30$ sufficient for statistically significant results, as confirmed by the exact permutation null on κ (Appendix B.1).

F Layer sweep projection figures

In the layer-sweep projections below, each figure shows one model, with a panel for each of the eight swept layers (titled by absolute layer index and relative depth). Large outlined dots are the pure-task centroids and small dots the individual samples; the grey $\times$ is the ratio-predicted mixed centroid and the outlined grey dot the actual mixed centroid, joined by a dotted line. Per-ratio κ values are reported in Appendix B.2 rather than on the plots. Colors identify the pure tasks as given in each figure’s legend.

We include two representative layer sweeps—Arithmetic and Entity for Llama 3.2 3B, the model used for the residual-ablation analysis; an MMLU example is shown in Figure 1. The complete set of projections for all five models and all three task groups is available in the accompanying code repository.

Arithmetic — Llama 3.2 3B: LDA projections of contrast vectors by layer

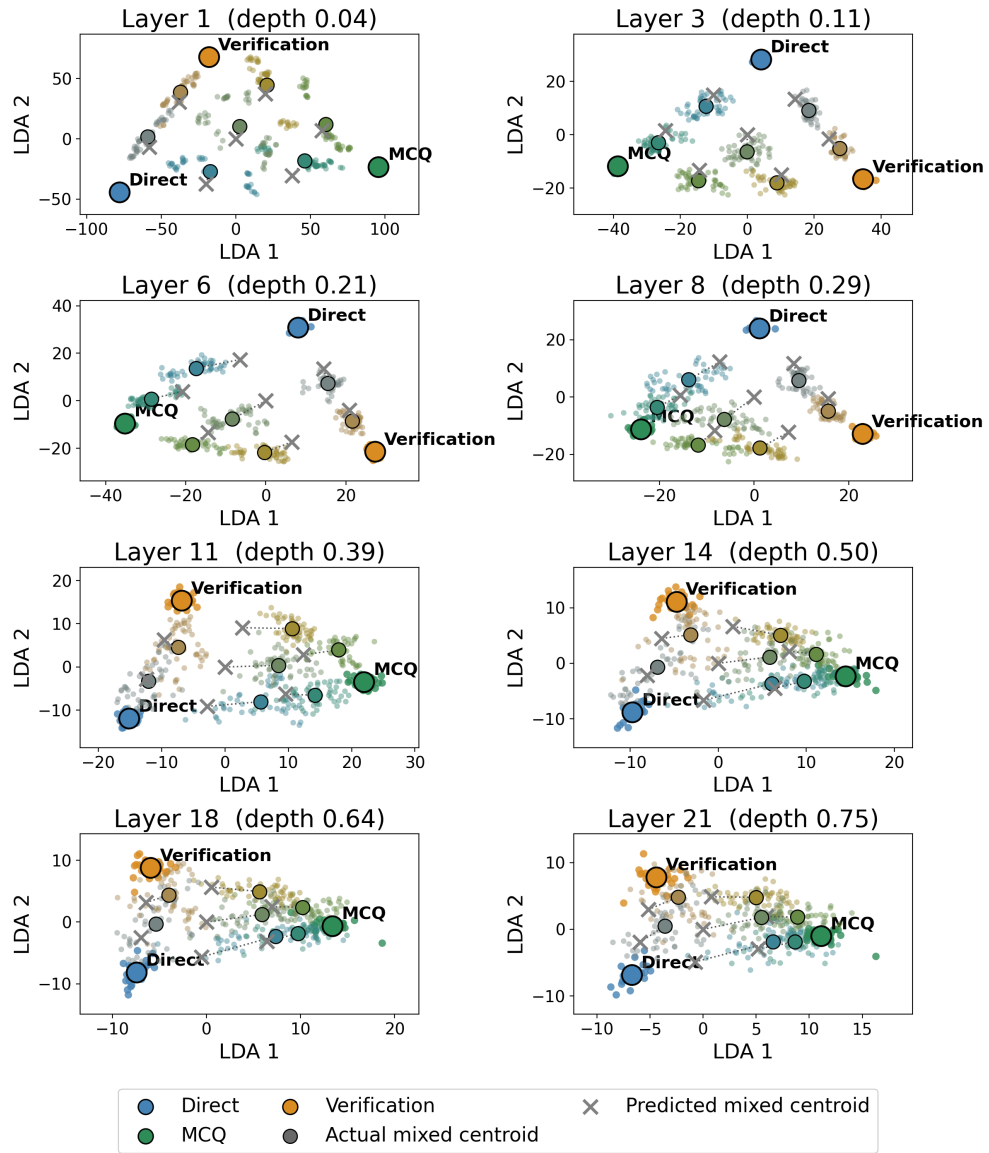


Figure 5: Arithmetic layer sweep projections — Llama 3.2 3B.

Entity — Llama 3.2 3B: LDA projections of contrast vectors by layer

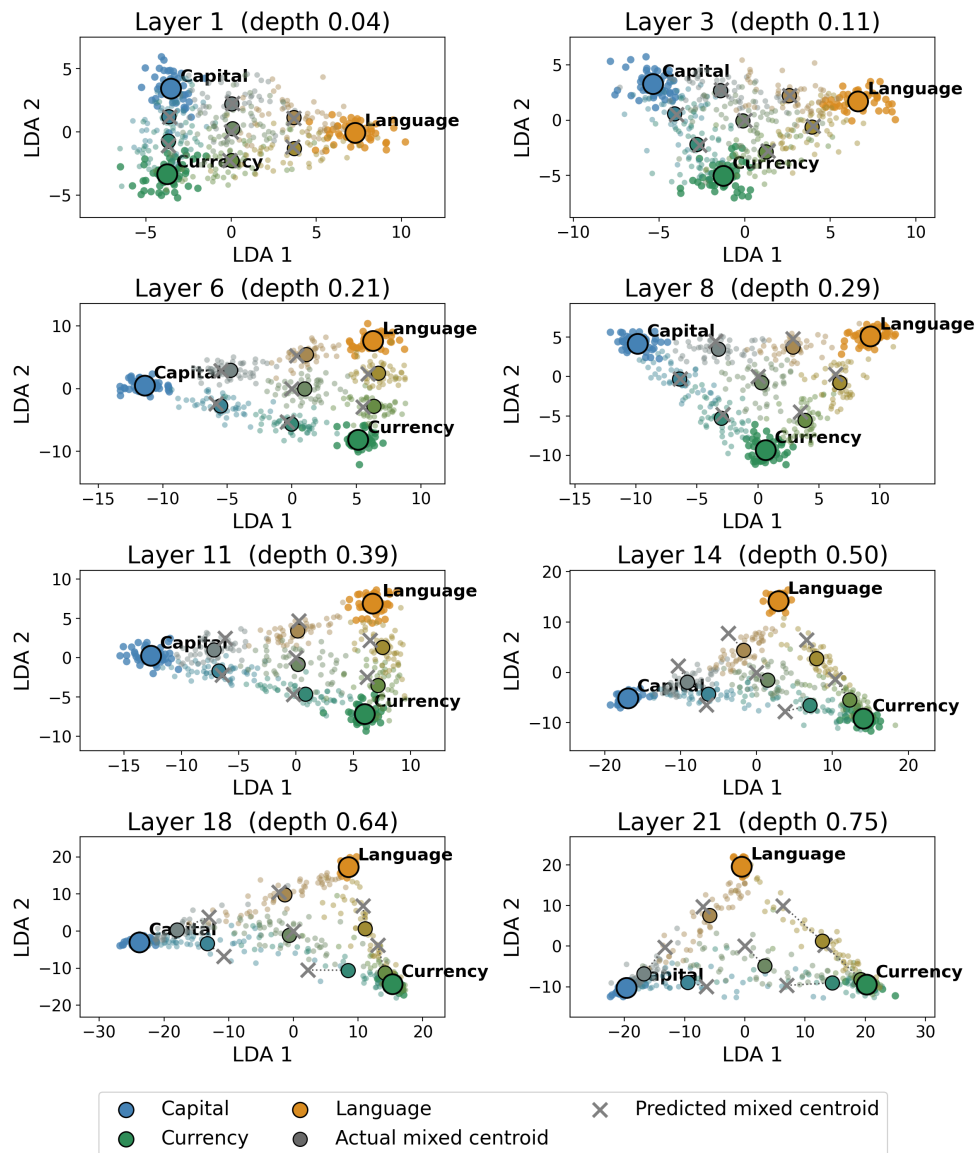


Figure 6: Entity layer sweep projections — Llama 3.2 3B.